

Loops

03 April 2026 05:08

There are two types of loops:

1. Entry Control Loops
 - i. while() loop
 - ii. for() loop
2. Exit Control Loop
 - i. do{...}while() loop

while loop

Syntax:

```
while(<condition>){  
    //code to repeat  
}
```

for loop

Syntax:

```
for(<initializations>;<condition>;<update>){  
    //code to repeat  
}
```

do...while() loop

Syntax:

```
do{  
    //code to repeat  
}while(<condition>);
```

1. WAP to input two numbers and find out quotient and remainder without using arithmetic operators(/, %).

Input: Enter first number: 45
Enter second number: 7

Output:

Quotient: 6
Remainder: 3

Concept of division is continuous subtraction, till the result is less than or equal to the divisor

```
45  
- 7  
---  
38  
- 7  
---  
31  
- 7  
---  
24  
- 7  
---  
17  
- 7  
---  
10  
- 7  
---  
3
```

6 ← quotient.

3 ← remainder

By default the condition is always true.

`for(;;){` ← This is the example of infinite loop.

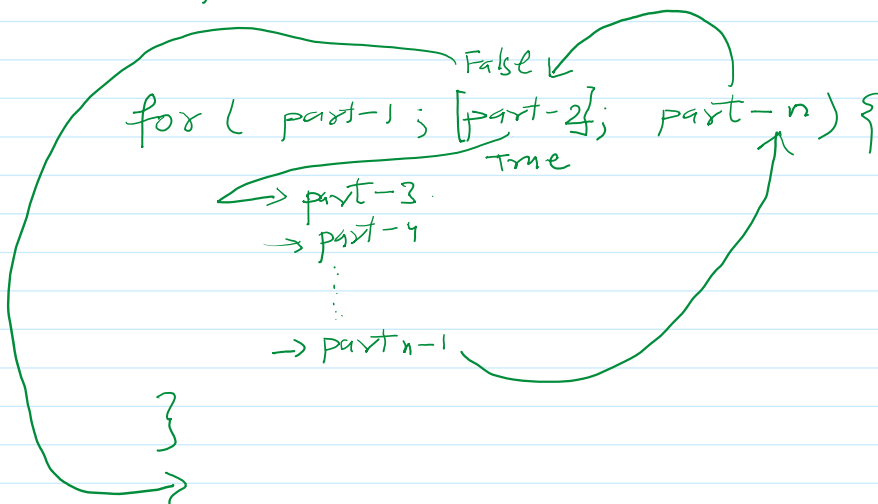
}

```
public class InfiniteForLoop
{
    public static void main(String[] args){
        System.out.print("\f");
        for(;;){
            System.out.print("\nHello!, I am Infinite loop");
        }
    }
}
```

optional

for (Any valid java statement ; < condition > ; Any valid java statement)
 {
 [Except, branching & looping statement]
 [Except, branching & looping statement]
 }
 // code to execute.
 }

This for loop is an advance form of while loop.



2. WAP to input a number and find number of digits and sum of digits.

Example:

Input: Enter any number: 3456

Output: Number of digits: 4

Output: Sum of digits: 18

$$(1+2)/(1*2) + (1+2+3) / (1*2*3) + (1+2+3+4) / (1*2*3*4) = 2.9167$$

package Iterations;

```

import java.util.*;

/**
 * Write a description of class Q8 here.
 *
 * @author (your name)
 * @version (a version number or a date)
 * (1+2)/(1*2) + (1+2+3) / (1*2*3)+ ...+(1+2+...+n)/(1*2*...*n) = result
 */
public class Q8
{
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        int i, num = 1, nsum = 1, pdeno = 1, n;
        double sum = 0;

        System.out.print("\nEnter the value of n: ");
        n = sc.nextInt();

        for(i = 1; i <= n-1; i++){
            num++;
            nsum += num;
            pdeno *= num;
            sum += nsum/(double)pdeno;
        }
        System.out.print("\nS = " + sum);
    }
}

```

```

package Iterations;
import java.util.*;

/**
 * Write a description of class Q10 here.
 * Frequency of each digit present in the number.
 * @author (your name)
 * @version (a version number or a date)
 */
public class Q10
{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        long num, countFreq = 0, i, t, d;
        System.out.print("\nEnter any positive integer value: ");
        num = sc.nextLong();

        for(i = 0; i < 10; i++){
            for(countFreq = 0, t = num; t > 0; t/=10){
                d = t % 10;
                if(d == i){
                    countFreq++;
                }
            }
        }
    }
}

```

```

        if(countFreq > 0){
            System.out.print("\nThe frequency of " + i + " = " + countFreq);
        }
    }
}

```

Output:

```

Blue: Terminal Window - Kapish11
Options
Enter any positive integer value: 3310092567184

The frequency of 0 = 2
The frequency of 1 = 2
The frequency of 2 = 1
The frequency of 3 = 2
The frequency of 4 = 1
The frequency of 5 = 1
The frequency of 6 = 1
The frequency of 7 = 1
The frequency of 8 = 1
The frequency of 9 = 1

```

```

public class Q9
{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        long t, num, newNum = 0, digit;
        System.out.print("\nEnter any positive integer: ");
        num = sc.nextLong();

        for(t = num; t > 0; t /= 10){
            digit = t % 10;
            if(digit > 0){
                newNum = newNum * 10 + digit;
            }
        }
    }
}

```

num : 5400207
 t : 5400207
 newNum: 7
 digit: 7

1, 1, 2, 3, 5, 8, 13, 21, 34, 55

```

10 public class P11
11 {
12     public static void main()
13     {
14         Scanner sc=new Scanner (System.in);
15         System.out.println("\nEnter the number");
16         int n=sc.nextInt() , a=1,b=0,f=0;
17         for(f<n;f++)
18         {
19             f=a+b;
20             if(f==n)
21             {
22                 System.out.println(n+ "is a part of the fibonacci sequence");
23                 System.exit(0);
24             }
25             a=b;
26             b=f;
27         }
28         System.out.println(n+ "is not a part of the fibonacci sequence");
29     }
30 }
31
32

```

n = 55
 a = 1, 1, 2, 3, 5, 8, 13, 21
 b = 1, 1, 2, 3, 5, 8, 13, 21, 34
 f = 1, 1, 2, 3, 5, 8, 13, 21, 34, 55

```
29 }  
30 System.out.println(n+ "is not a part of the fibonacci sequence");  
31 }  
32 }  
Class compiled successfully
```